🌿 Project Overview & Mission

Core Identity

NutriSmart AI combines evidence-based nutritional science, AI biometric meal personalization, and visual portion cheat-sheets to make healthy eating effortless, sustainable, and enjoyable. Features full light/dark mode and mobile responsiveness.

1 Page 1: Home (Welcome & Visual Spotlight)

Route: 'home'

The primary entry point that immediately sets the clean, health-focused tone of the brand.

Left Column (Value Proposition):
Main title: *"Personalized Nutrition for a Healthier, Balanced Life"* with intro text, CTA buttons (*Explore Recipes*, *AI Meal Planner*), and 3 benefit checkmarks.

Right Column (Visual Salad Bowl):
High-definition colorful salad bowl photo with floating ambient glow and interactive badges: *"🥗 Fresh & Wholesome"* & *"⚡ Balanced Nutrition"*.

3 Feature Highlight Cards:
Wholesome Recipes (nutrient-dense real food), *Daily Nutrition Tips* (science-backed guidance), and *Personalized Wellness* (mindful vitality).

Daily Eating Tip & Recipe Preview:
Interactive wisdom card with copy/save action + preview cards of wholesome recipes with calories and prep times.

2 Page 2: Daily Tips (Nutritional Wisdom Archive)

Route: 'tips'

Daily micro-habits grounded in metabolic science and clinical research to build sustainable vitality.

Interactive Real-Time Search:
Search bar allowing instant filtering by keywords like *protein, hydration, gut, breakfast, sleep*.

Category Navigation Pills:
Quick filters: *Gut Microbiome & Immunity*, *Metabolism & Energy*, *Hydration*, and *Mindful Habits*.

Actionable Tip Cards:
Includes headline, explanation, *Science Insight* callout, and a clear
Action Step
to implement today.

Save & Share Controls:
One-click clipboard copy and local browser storage to bookmark favorite daily habits.

3 Page 3: Healthy Recipes (Wholesome Recipe Hub)

Route: 'recipes'

A rich catalog of whole-food recipes designed for balanced macros, great taste, and quick home prep.

Filter by Goal & Diet:
Filter categories: *High Protein*, *Vegetarian*, *Low Carb*, *Heart Healthy* + tags for *Gluten-Free, Keto, Vegan*.

Nutritional Badges:
Every recipe displays exact Calories (kcal), Protein (g), Carbs (g), Healthy Fats (g), Fiber (g), and Cook Time.

Recipe Detail Pop-up Modal:
Clicking any card opens a clean modal with full ingredients checklist, step-by-step instructions, and chef tips.

AI Score Indicator:
Wholesomeness rating (e.g., 98/100) based on real-food ingredient density and absence of ultra-processed additives.

4 Page 4: AI Meal Planner (Personalized Nutrition Engine)

Route: 'planner'

Algorithmic meal schedule generator customized to user biometrics and lifestyle targets.

Biometric Profile Form:
Captures age, gender, height, weight, activity level multiplier, and health goal (*Weight Loss, Muscle Building, Maintenance*).

Dietary & Allergy Filters:
Select preferences: *Mediterranean*, *High Protein*, *Keto*, *Vegetarian*, or *Dairy-Free*.

Personalized Day & Weekly Plan:
Calculates targeted calories and macros, then structures Breakfast, Lunch, Dinner, and Snack recommendations.

Automated Grocery List:
Aggregates ingredients across all generated meals into a printable, categorized weekly shopping checklist.

5 Page 5: Nutrition Tools (Calculators & Macro Distribution)

Route: 'tools'

Fast, scientific calculation tools for personal body metrics and macronutrient balancing.

Interactive BMI Calculator:
Calculates Body Mass Index with color-coded classification: Underweight, Normal (18.5–24.9), Overweight, Obese.

BMR & TDEE Target Estimator:
Computes Basal Metabolic Rate and Total Daily Energy Expenditure based on physical activity multipliers.

4 Macro Distribution Boxes:
Visual breakdown for *Proteins (4 kcal/g)*, *Carbohydrates (4 kcal/g)*, *Healthy Fats (9 kcal/g)*, and *Fiber*.

Goal Presets:
One-click macro recalibration for Fat Loss (high protein), Muscle Hypertrophy, or Longevity/Maintenance.

6 Page 6: Nutrition Guide (Balanced Diet & Knowledge Hub)

Route: 'guide'

Comprehensive visual education hub highlighting balanced diet foundations and interactive charts.

Why a Balanced Diet Matters:
4 pillars: *All-Day Energy* (no crashes), *Immune Defense*, *Gut & Serotonin*, and *Metabolic Longevity*.

The Sustainable 80/20 Rule:
80% whole nutrient-dense foods (vegetables, proteins, whole grains), 20% culinary flexibility without guilt.

Search Bar & 10 Topic Cards:
Real-time search + filters (Balanced Diet, Macros, Micros, Gut, Timing). Cards have summaries, 💡 Smart Rules, & sources.

50–25–25 Smart Plate Method:
Interactive plate diagram: 50% Veggies & Fruits, 25% Lean Protein, 25% Complex Carbs, plus hand portion rules.

Food Label Inspector (5/20 Rule):
Clickable pins decoding serving sizes, calories per serving, hidden sugars, and the 5% (Low) / 20% (High) DV metric.

Eating Day Timeline:
Chronobiological guide from 7:30 AM morning protein pacing to 7:30 PM screen-free mindful dinner.

🎤 3-Point Presentation Script (What to Say to Evaluators)

1. The Hook:	*"NutriSmart AI solves diet confusion by pairing biometric AI planning with simple visual rules like the 50–25–25 plate and 80/20 balance."*
2. Technical Flow:	*"Built using React 19 and Vite with modular components, localStorage state persistence, instant search, and dynamic responsive CSS."*
3. Key Impact:	*"Empowers users across 6 structured pages: from daily tips and macro calculation to personalized meal plans and label literacy."*